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INVITED REVIEW

A systematic review of the internet of things and artificial intelligence applications in milk quality monitoring and analysis

RAHUL MHAPSEKAR,[1,*] DEIRDRE KILBANE,[1] STEVEN DAVY,[2] LIZY ABRAHAM,[1] MARK FENELON[3] and NORAH O'SHEA[3]

[1]*Walton Institute, SETU, South East Technological University (SETU) West Campus, Carriganore, Waterford X91 P20H, Ireland,* [2]*Centre for Sustainable Digital Technologies, Technological University Dublin, Dublin, Ireland, and* [3]*Food Chemistry and Technology department, Teagasc Food Research Centre, Teagasc, Fermoy, Co. Cork, Cork P61 C996, Ireland*

Background:
Milk quality analysis (MQA) is critical for ensuring high-quality dairy production and enhancing the productivity of dairy processing plants. The integration of Internet of Things (IoT) and artificial intelligence (AI) can revolutionise MQA by enabling automated, real-time detection and monitoring of key parameters (i.e. composition, adulterants, indicators of microbial quality), reducing human error and improving decision-making in the dairy supply chain.

Aim(s):
This systematic review aimed to evaluate the application of IoT and AI in MQA over the past decade, focussing on three critical areas: composition monitoring, adulteration detection and indicators of microbial quality.

Methods:
A systematic analysis of 43 studies was conducted, comparing sensing technologies (e.g. spectroscopy and gas sensors), data communication protocols (i.e. WifI, Bluetooth), data processing strategies (edge computing, cloud platforms) and AI models (Machine Learning, Deep Learning). The methodologies implemented, along with model performance evaluation metrics and the scalability of the proposed technologies, will be assessed.

Major Findings:
Spectroscopy emerged as a leading method for compositional and adulteration analysis (62%), while gas sensors were preferred for microbial spoilage detection. Edge computing was the leading data processing strategy (64%), enabling real-time analysis, though hybrid edge–cloud platforms demonstrated good scalability potential. Machine Learning models (e.g. Support Vector Machines, Random Forest) achieved high predictive accuracy, while Deep Learning excelled in classification tasks, particularly for adulteration detection.

Scientific Implications:
Despite the significant advancements, challenges remain in developing generalised models applicable across diverse milk samples (geographic regions and milk species). Current systems face limitations in data communication from remote farms with poor connectivity and computational constraints when processing complex spectral data on edge devices. Future advancements in 5G networks could address latency issues in large-scale deployments, while smart sensors with embedded preprocessing capabilities may reduce computational constraints. Hybrid edge-cloud platforms enable real-time analysis while supporting model updates, enhancing smart dairy monitoring.

Keywords Artificial intelligence, Internet of things, Microbial quality monitoring, Milk adulteration detection, Milk quality analysis, Systematic literature review.

*Author for correspondence. E-mail: rahul.mhapsekar@waltoninstitute.ie

INTRODUCTION

To maintain the quality assurance of milk, it is essential to monitor the physicochemical and microbial properties, since they can impact the nutritional and processing performance of milk (Zhuozhuo and Guo 2021). Key milk components, such as fat, protein and lactose, influence pricing, while Total Plate Count (TPC) and Somatic Cell Count (SCC) are important milk quality indicators (Nousiainen *et al*. 2007), but these two areas are not the focus of this review.

Milk adulteration remains a global challenge despite regulatory measures in Ireland and Europe affecting milk quality. Common adulterants include water, vegetable oil and detergents, added to increase volume (Azad and Ahmed 2016). In extreme cases, such as the Chinese melamine scandal, harmful substances were used to artificially raise protein content (Gossner *et al*. 2009). This highlights the urgent need for rapid, robust and cost-effective adulteration detection methods. Current approaches are often slow and expensive, requiring trained personnel, thereby creating a demand for faster, accessible alternatives such as spectroscopy and smart technologies (Azad and Ahmed 2016; Natarajan and Ponusamy 2021).

Microbial activity in milk poses risks to both quality and safety. Pathogens such as *Listeria monocytogenes*, *Salmonella* spp., *Yersinia enterocolitica* and *Staphylococcus aureus* are known to cause foodborne illness, while *Pseudomonas fragi* and *Lactobacillus* impact product quality (Vasavada 1993). Integrating technologies such as the Internet of Things (IoT) and artificial intelligence (AI) enables rapid, accurate and efficient milk quality analysis (MQA).

The IoT framework can be divided into three sections: the object end, the network and the cloud. Sensors and other smart devices operate in the object end; the cloud is used for data storage, and the communication between the two is established using the network end (Panikuttira *et al*. 2018) where several communication protocols such as Bluetooth Low Energy (BLE), Wi-Fi, Global Systems for Mobile Communication (GSM) and ZigBee are used to transfer data among these blocks. Defining these functional blocks helps optimise platform design (Ray 2018). The IoT architecture is shown in Figure 1; the major functional blocks are as follows:

a Physical Layer: IoT devices such as sensors, actuators, processing units, controllers and display devices (Figure 1a).
b Network Layer: It enables communication between IoT devices and the data servers or cloud. Several communication protocols operate in data link, network, transport and application layers (Figure 1b).
c Service Layer: It enables technologies required for data processing. It involves data management, data visualisation, data analytics and management of connected devices (Figure 1c).
d Application Layer: It enables users to centrally manage the IoT system and provides intuitive data visualisations to support informed decision-making by end users. It acts as an interface between users and the IoT system itself (Figure 1d).

AI integrated with IoT enhances MQA by analysing milk production data, such as fat and protein levels, and detecting adulteration. This helps dairy plants work more efficiently by providing real-time milk quality reports for decision-making. For instance, if the AI system detects changes in composition, the process manager can inform standardisation processes for different applications, that is cheese production, protein fortified products quality and economic return through appropriate segregation within the plant.

Machine Learning (ML), a subfield of AI, uses advanced algorithms to solve complex problems that traditional methods cannot address effectively (Slob *et al*. 2020). Milk spectral data is often nonlinear and does not always follow assumptions of normality. Machine Learning algorithms can help extract valuable insights from this spectral data to analyse milk composition. Compared to traditional linear models, ML models offer robust solutions for data containing outliers and missing values. They can also capture nonlinear relationships between observations and required classes, increasing the accuracy and robustness of classification-related problems in MQA (Zhang *et al*. 2021).

An ML prediction model is created by training an algorithm on a dataset containing features (independent variables) and outcomes (dependent variables). The trained model is then validated using a separate dataset to assess its performance. If the model performs well on validation, it can be applied in practice (Slob *et al*. 2020). While the ML process involves training, testing and application, challenges arise in selecting features, choosing algorithms and handling large datasets effectively. Thus, it is necessary to select the best data pre-processing techniques prior to processing the milk quality data through an AI algorithm. Pre-processing is essential for transforming raw data into meaningful inputs, improving the accuracy and robustness of AI/ML models (Al-jabery *et al*. 2020). In ML, the use of proper evaluation parameters (accuracy, precision) is critical for achieving optimised results. Without rigorous evaluation, the outcomes of the model cannot be trusted or validated, making the results meaningless and unreliable (Reich and Barai 1999). Thus, the data pre-processing techniques and selection of appropriate evaluation methods are of utmost importance for achieving accurate results in applications such as MQA.

A Systematic Literature Review (SLR) uses a structured method to identify gaps and guide future research. Existing SLRs cover AI/ML in dairy contexts—such as livestock monitoring (García *et al*. 2020), farm management (Kassahun *et al*. 2022), cattle identification (Hossain *et al*. 2022), and

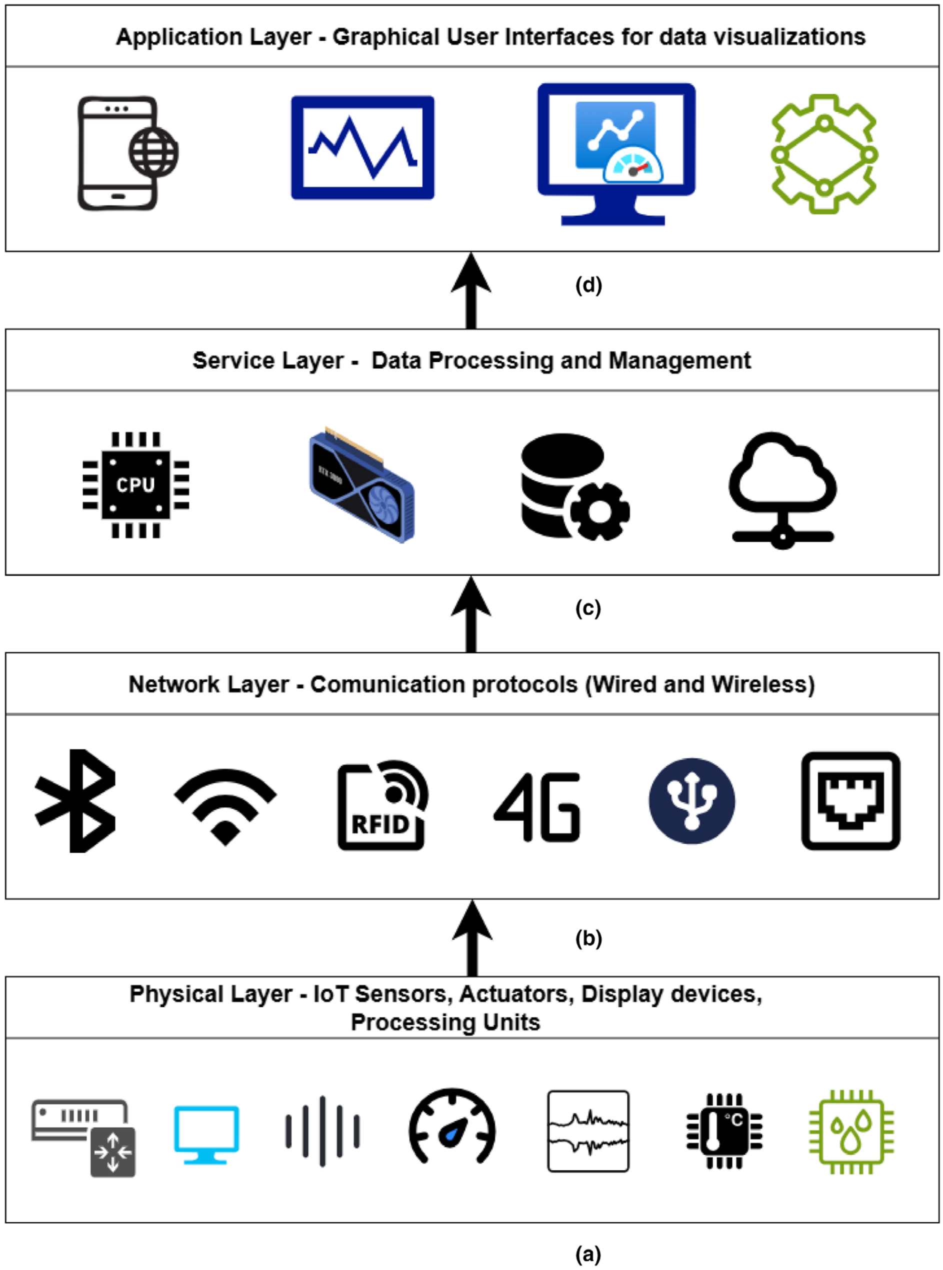


Figure 1 Internet of things architecture - (a) Physical Layer, (b) Network Layer, (c) Service Layer, (d) Application Layer.

precision agriculture (Cisternas *et al*. 2020). Others examine IoT and AI in broader food quality and safety domains (Rejeb *et al*. 2022; Chakraborty *et al*. 2023; Hasan *et al*. 2023).

Natarajan and Ponusamy (2021) reviewed detection methods such as FTIR, NIR and Raman Spectroscopy, alongside chromatographic techniques, noting their inefficiency for real-time use. They advocate for rapid, real-time solutions. Slob *et al*. (2020) discussed ML in dairy farm management, highlighting Artificial Neural Networks (ANNs) for tasks such as disease and quality prediction, with superior performance over traditional models (Dongre and Gandhi 2016).

Mahmud *et al*. (2021) explored Deep Learning (DL) in precision cattle farming, particularly Convolutional Neural Networks (CNNs) for health and behaviour monitoring, noting challenges in data acquisition and processing. With increasing data availability, Cabrera and Fadul-Pacheco (2021) stressed the role of AI/IoT in integrating systems (e.g. herd software, milking records) to predict diseases such as mastitis and improve efficiency.

Goyal *et al*. (2022) focussed on real-time adulteration detection, emphasising the potential of DL models and providing open-source datasets. However, issues such as high costs, data security and system integration remain unresolved. Despite these studies, no comprehensive SLR specifically addresses IoT and AI/ML applications in MQA. This review aimed to fill that gap, focussing on milk composition, adulteration detection and microbial analysis.

METHODOLOGY

The SLR methodology employed in this study follows the guidelines outlined by Kitchenham and Charters (2007) and consists of three main phases: Planning, Conducting the Review and Reporting the Results. During the planning phase, research questions and a search strategy were defined, which included selecting relevant databases, creating search strings and establishing selection criteria. The review phase involved executing the search strategy to gather relevant publications, followed by a detailed evaluation of papers meeting the selection criteria. The reporting phase addressed the research questions through analysis presented in figures and tables.

The research objectives investigate key challenges in MQA prior to processing, identify IoT technologies used for smart solutions, explore AI applications in MQA, evaluate AI/ML algorithms employed and determine the best-performing algorithms and their evaluation metrics. To comprehensively address these research objectives, a search was conducted across databases including IEEE Xplore, Science Direct, Springer, Wiley, Web of Science and Scopus, supplemented by snowballing, ResearchGate and Google Scholar. Snowballing involved identifying related works by examining references in selected articles, while ResearchGate was used to access studies not available via open access.

Search strings were tailored to each database using combinations of keywords such as ‘Dairy’, ‘Milk Quality’, ‘Internet of Things’, ‘Artificial Intelligence’, ‘Machine Learning’ and ‘Deep Learning’, applied across titles, abstracts and keywords where possible. Broader terms were employed for databases with limited search functionalities. The inclusion criteria focussed on publications addressing IoT and AI solutions for MQA, while exclusions were applied to non-English papers, duplicates, pre-2012 publications, unrelated studies and reviews. This methodology ensured a rigorous and comprehensive review of relevant literature, providing valuable insights into the role of IoT and AI in advancing MQA practices.

RESULTS

This section addresses the research objectives using data from 43 papers on MQA. The studies were categorised into three groups: IoT-focussed (10 papers), which implemented hardware across system layers without AI/ML. The second category included AI/ML-focussed (10 papers), which used algorithms on existing datasets without hardware components, and hybrid studies (23 papers), which integrated hardware prototypes with AI/ML models to improve prediction and decision-making. For a more comprehensive SLR, we combined 33 IoT-focussed and 33 AI/ML-focussed studies, accounting for the 23 overlapping hybrid studies. DL was included under the broader AI category. The research further addresses three core aspects of milk quality: composition monitoring (51% of studies), adulteration detection (42%) and microbial quality indicators (7%).

Milk composition monitoring: IoT and AI technologies can revolutionise milk composition monitoring by integrating various sensors, such as spectral, electrochemical, pH, temperature and humidity sensors, to continuously monitor and collect data on milk quality. IoT devices transmit this real-time data to a central system, where AI algorithms analyse it to identify patterns, predict composition and identify safety issues. Fat, total protein, lactose, casein and Solid-Not-Fat (SNF) were measured and detected as a part of milk composition analysis.

Most studies that included fat measurements had a detection threshold between 3.5 and 4.5%. Milk profile (whole, semiskimmed or skimmed milk) detection was also investigated in one of the studies, and according to Ali *et al*. (2020), the fat content in milk is indicated as follows: 3.5% for whole milk, 1.5–1.8% for semiskimmed milk, and not greater than 0.5% for skimmed milk.

Adulteration detection: The frequently reported adulteration parameters included pH, temperature, gases, salinity and odour. These parameters can be used as indicators for milk quality and microbial spoilage. The following studies (Neto *et al*. 2019; Alagumeenaakshi *et al*. 2021; Natarajan and Vijayakumar 2021; Nayeem *et al*. 2021; Saravanan *et al*. 2021; Kumar *et al*. 2022) observed that milk is adulterated to falsely increase the milk volumes by the addition of water, detergents, soap, chemicals, urea, caustic soda, formaldehyde and melamine. Adulterants such as sulfate salts, cane sugar, starch, urea and common salts can falsely increase the SNF value of milk. Milk adulteration can lead to changes in its nutritional composition and, in certain cases, pose health risks to consumers. For instance, adulterants such as peroxide and detergents can cause gastritis. Similarly, excessive levels of starch should be avoided by diabetic patients, and urea consumption can cause kidney damage in humans (Azad and Ahmed 2016).

Microbial quality indicators: The growth of certain bacteria can cause milk spoilage, making it unfit for human consumption. Some bacteria mentioned in studies (Asif *et al*. 2016; Ady *et al*. 2018; Kadam and Shinde 2018) are as follows: *Listeria monocytogenes*, *Salmonella* spp., *Staphylococcus aureus*, *Pseudomonas fragi* and *Lactobacillus*. It is necessary to detect microbial activity to ensure high-quality and consistent milk for determining the shelf life of milk.

Microbial activity can lead to the production of acetic acid, ethanol and acetaldehyde, which can be detected using gas sensors (Asif *et al*. 2016, Ady *et al*. 2018, Kadam and Shinde 2018). Gases such as acetaldehyde, dimethyl sulphide and ethanol were measured in milk to evaluate microbial activity, along with changes in the smell and colour of milk. The sensors used to measure these gases were TGS (822, 813, and 2620). TGS sensors operate on the detection principle based on the chemical absorption and description of the gases on the surface of the sensor. When milk is kept at an unsuitable temperature for certain time periods, it accelerates bacterial growth, thus leading to faster spoilage. This highlights the importance of temperature monitoring.

Technologies applied at every layer for IoT deployment in MQA

The subsequent sections discuss the current state of the art technologies used for MQA using IoT systems. Table 1 indicates all the IoT components used in the studies included in this SLR. Figure 2 shows the IoT components used in the different layers of an IoT architecture in the studies reported in this SLR.

Physical layer

Sensors: Sensors are used for monitoring milk quality, acting as the core data collection element in IoT systems (Poghossian *et al*. 2019). From the reviewed papers in this SLR, it was noted that spectroscopy was predominantly utilised for MQA due to its ability to quickly evaluate essential milk components such as fat, protein, lactose and total solids. Spectroscopy appeared in 27 of the 43 studies analysed, with diverse spectral ranges tailored to distinct analytical requirements. Among these techniques, Near-Infrared Spectroscopy (NIRS) was the most frequently employed (six studies), followed by Visible–Near Infrared (VIS–NIR) (five studies), Mid-Infrared (MIR) (four studies) and Infrared (IR) spectroscopy (four studies). However, methods such as Fourier Transform Infrared (FTIR), UV–VIS and Visible Infrared (VIS) spectroscopy were less prevalent as the measuring principle for the sensing technology, having featured in three or fewer studies. Notably, one study utilised Raman spectroscopy to monitor milk processing, detecting deviations in fat content, added water, cleaning solutions and temperature variations through advanced spectral analysis (including Principal Component Analysis, Gaussian Process Regression, Autoencoders and other ML-based techniques). Spectroscopy played a significant role in analysing milk composition, with spectrometers utilised in 15 out of 22 related studies to perform milk quality analysis (Diaz-Olivares *et al*. 2020, Muñiz *et al*. 2020, Yang *et al*. 2020a, 2020b, Prasanth *et al*. 2021, Muhammad *et al*. 2019, Villar *et al*. 2012, Deshpande *et al*. 2021, Tao *et al*. 2022, Vimalajeewa *et al*. 2018, Dina *et al*. 2019, Bogomolov *et al*. 2012, Bogomolov and Melenteva 2013, Dixon *et al*. 2017, Ali *et al*. 2020). These instruments allowed for quick, nondestructive testing, demonstrating their importance in practical MQA applications. Besides spectroscopy, other sensors were used for more targeted applications, such as Light Detecting Resistors (LDRs) and lactometers, which were implemented in four studies to determine fat levels, Corrected Lactometer Reading (CLR) values and SNF content (Yadav *et al*. 2013; Neeraj *et al*. 2017; Drekalović 2021; Patil *et al*. 2021). One study employed the C12880MA spectrometer, operating in the 340–850 nm range, to show the difference between full fat and skim fat milk (Ali *et al*. 2020). Temperature sensors, including the low-cost DSB120, were used to track storage conditions, underscoring their significance in milk quality monitoring (Ma *et al*. 2018). For detecting milk adulteration, sensors focussing on pH, temperature, odour and salinity were broadly utilised. The DHT11 sensor for pH monitoring and the DSB1820 temperature sensor were most commonly employed. These sensors proficiently detected adulterants such as water, soap, and detergents. Gas sensors (e.g. TGS 2602, 2620 and 815) were employed in three studies to sense chemical solvents or gases that suggest adulteration (Tohidi *et al*. 2018; Anand *et al*. 2019; Nayeem *et al*. 2021). Furthermore, 12 studies showcased the use of spectrometers to identify compositional changes induced by adulteration. In microbial quality indicators, gas sensors such as TGS 813, 822 and 2620 played a key role in identifying gases related to spoilage. An innovative use of a magnetoelastic sensor was observed for detecting *S. aureus*, demonstrating the potential of advanced biosensors in milk quality monitoring systems (Asif *et al*. 2016; Ady *et al*. 2018; Kadam and Shinde 2018).

The frequent use of spectroscopy underscores its dependability and versatility in MQA, yet its application in real-time IoT systems may face obstacles due to high costs and system complexity. Future research should investigate cost-efficient miniaturised spectrometers or develop alternatives using optical or electronic principles to achieve comparable outcomes. Miniaturised sensors achieve a lower signal-to-noise ratio and reduction in resolution. Although gas sensors are effective for pinpointing microbial activity or adulteration, they are often application-specific and necessitate calibration for different scenarios. Integrating gas sensing with spectroscopy or other sensors could enhance robustness and yield a more thorough analysis. Interestingly, the restricted use of biosensors such as magnetoelastic sensors suggests a potential area for further research, particularly for specific microbial spoilage identification. Advancements in this field could greatly improve the accuracy of detecting microbial quality indicators in milk. Additionally, embedding multi-sensor systems within edge-centric IoT structures could further optimise real-time milk quality monitoring and adulteration detection, boosting the scalability and practicality of these systems.

Table 1 IoT components used in the studies for MQA.

Study	Physical layer sensors	Data acquisition system	Data processing unit	IoT platform	Communication protocol
Nayeem *et al*. (2021)	MQ3, DHT11, pH sensor, LDR sensor	Arduino	NodeMCU	BlynkIoT	Wi-Fi ESP8266
Saravanan *et al*. (2021)	HX711 weight sensor, pH sensor, Temperature sensor—DS18B20, Lactometer, colour sensor	NodeMCU	NodeMCU	Google Firebase	Wi-Fi ESP8266
Alagumeenaakshi *et al*. (2021)	pH sensor, ultrasonic sensor, colour sensor, viscosity sensor	Atmega328	PC	ThingSpeak	Ethernet
Haribabu *et al*. (2019)	Lactometer, pH sensor	Raspberry Pi3	Raspberry Pi	ThingSpeak	Wi-Fi, GSM, and RFID for farmer identification
Drekalović (2021)	Colour sensor, turbidity sensor, pH sensor, temperature sensor	Smart Can—Includes sensors and processor inbuilt	Cloud	Custom	GSM, BLE, Wi-Fi
Neeraj *et al*. (2017)	IR sensor, Lactometer	Arduino	Arduino	ThingSpeak	Ethernet
Yadav *et al*. (2013)	MilkoTester	MilkoTester	MilkoTester	N/A	RFID
Ali *et al*. (2020)	C12880MA spectrometer	Arduino	Raspberry Pi	Custom	N/A
Kadam and Shinde (2018)	TGS 813, 822, 826, 2620	Arduino	Cloud	BlynkIoT	Wi-Fi ESP8266
Asif *et al*. (2016)	TGS 2620, 813, 822	LPC2148	PC	Custom	ZigBee
Natarajan and Vijayakumar (2021)	Spectral sensor—AS72651, AS72652, AS72653	Arduino	PC	Custom	Wi-Fi ESP8266
Anand *et al*. (2019)	TGS2620, 815, 2602	MSP430	PC	Custom	USB
Moharkar and Patnaik (2019)	IR sensor—TSL2561	Raspberry Pi	Raspberry Pi	Custom	Bluetooth, USB, Ethernet
Pablo *et al*. (2020)	Biosensor, Ocean optics STS VIS–NIR spectrometer	Sensor	PC	Custom	Serial Interface
Farah *et al*. (2021)	Differential scanning calorimetry sensor	Sensor	PC	Custom	Not Specified
Hosseini *et al*. (2021)	FT-NIR map spectrometer	Sensor	PC	Custom	Not Specified
Vasafi *et al*. (2018)	INNO-Spec Raman 785 spectrometer	Sensor	PC	Custom	USB
Tohidi *et al*. (2018)	Gas sensors—MQ3, MQ8, MQ137, TGS 813, 822, 2602, 2620	Data acquisition card – N1USB6009	PC	Custom	USB
Lal *et al*. (2022)	Temperature—DSB1820, pH &EC analog v2 sensors by gravity	Raspberry Pi	PC	ThingSpeak	Wi-Fi ESP8266
Vallejo *et al*. (2022)	WPT coil sensor	Vector network analyser	PC	Custom	Not Specified
Patil *et al*. (2021)	LDR	Arduino	Raspberry Pi	Custom	Ethernet
Fanglin *et al*. (2020)	TGS 2600, 2611, 822, 832, 2620, 2602	Arduino	PC	Custom	USB
Diaz-Olivares *et al*. (2020)	NIR spectrometer	Optical measuring equipment	PC	Custom	Serial Interface
Ma *et al*. (2018)	Temperature Sensor—DS18B20	MSP430	Cloud	AWS	Wi-Fi, GSM SIM900A for users
Muñiz *et al*. (2020)	NIR spectrometer—Microphazir 1624	Sensor	Cloud	Custom	USB, Wi-Fi
Yang *et al*. (2020b)	NIR sensor	Sensor system	Raspberry Pi	N/A	Wi-Fi

(continued)

Table 1 (Continued).

Study	Physical layer sensors	Data acquisition system	Data processing unit	IoT platform	Communication protocol
Yang *et al.* (2020a)	VIS–NIR multispectral sensor AS7265x, Milkoscan	Sensor system	Raspberry Pi	Custom	USB
Prasanth *et al.* (2021)	NIR spectrometer AS7262, AS7263	Sensor system	Raspberry Pi	N/A	Serial interface – i^2C, UART
Muhammad *et al.* (2019)	Monochromator sensor	Camera	Raspberry Pi	Custom	USB
Villar *et al.* (2012)	VIS–NIR sensor	Sensor system	PC	Custom	Not Specified
Deshpande *et al.* (2021)	NIR sensor—AS7263	Arduino	PC	Custom	i^2C, USB
Tao *et al.* (2022)	NIR sensor—AS7263	Data acquisition card	PC	Custom	Serial Interface
Ady *et al.* (2018)	MQ136, 137, TGS2602	Arduino	PC	Custom	USB

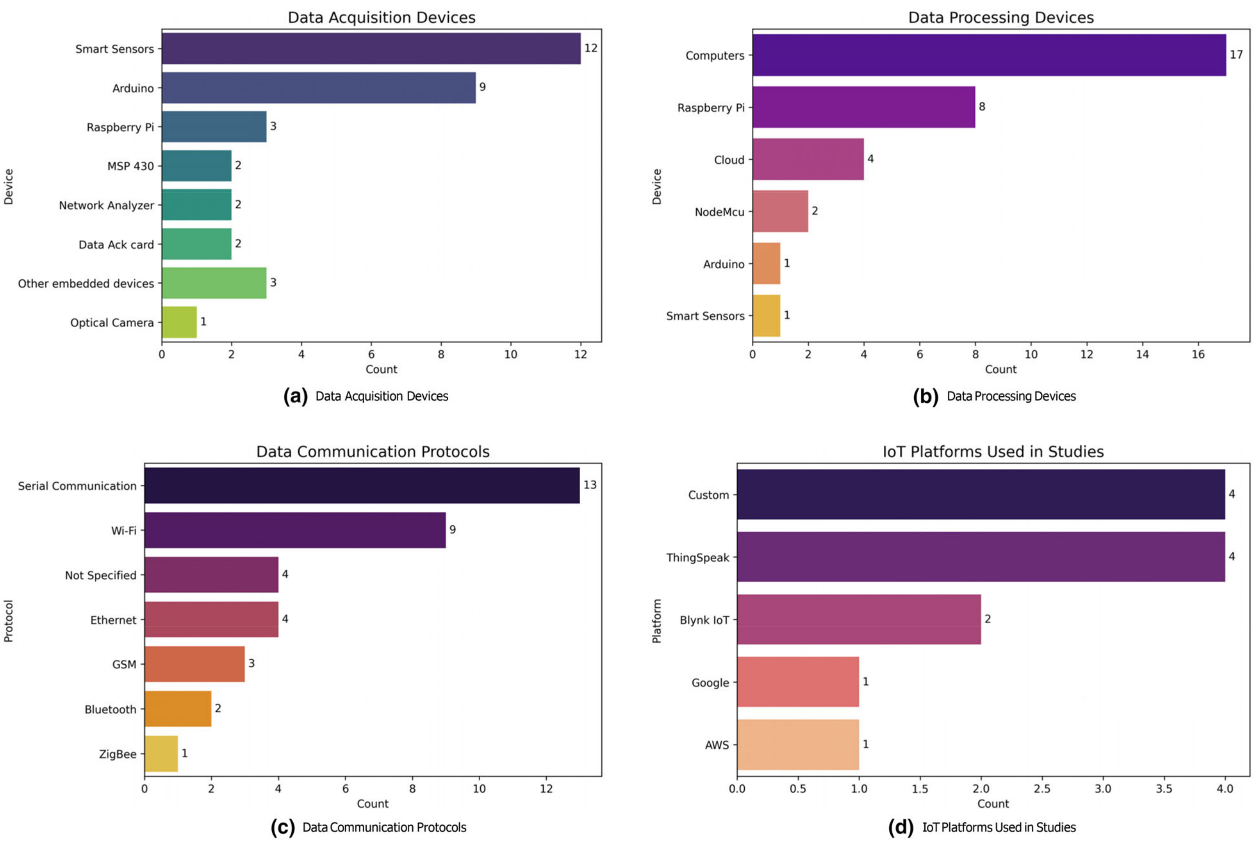


Figure 2 Distribution of devices, IoT platforms and communication protocols used in studies - (a)Data Acquisition Devices, (b) Data Processing Devices, (c) Data Communication Protocols, (d) IoT platforms.

Data processing units: In MQA studies, data processing units play a crucial role in transforming sensor data into practical insights. These units vary from edge devices to cloud computing systems, facilitating both real-time and remote analyses. Of the 43 studies reviewed, 33 involved hardware implementations, with details outlined in Table 1. Findings from these studies emphasise the dynamic role of data acquisition and processing in current MQA systems. Notably, 33% of the studies employed smart sensor systems with integrated data acquisition, real-time milk quality monitoring and adulteration detection, improving the scalability and practicality of these systems. The type of data acquisition units used in the reported studies are shown in Figure 2(a,b).

Among edge devices, the open-source Arduino hardware prototyping board was frequently used to connect with milk quality sensors for data acquisition. Other platforms such as ARM MSP430, network analysers, NodeMCU, LPC2148, Atmega328 and optical cameras comprised 30% of the studies, reflecting diverse methods for acquiring sensor data. A central aspect of these studies is the processing location—either locally on edge devices or remotely in the cloud. Most studies (17 out of 33) processed sensor data using on-site Personal Computers (PCs) or laptops, indicating a dependence on traditional computing systems. Nonetheless, edge computing is gaining popularity, with Raspberry Pi being the most frequently used edge processor as seen in eight studies. NodeMCU was employed twice, whereas Arduino and smart sensor systems each appeared in a single study. In four studies, cloud-based processing was utilised, demonstrating its capacity for managing substantial data volumes or for IoT ecosystem integration.

The dominance of traditional PCs and laptops for on-site processing reflects a preference for their computational power and familiarity. However, this approach limits portability and scalability, especially for rural or distributed dairy farms. The increasing use of edge devices such as Raspberry Pi signals a shift towards more compact, cost-effective and energy-efficient solutions. These devices not only reduce latency (i.e. the delay in transmission and processing of milk quality data, which can slow real-time detection of adulteration) but also align with the growing demand for Edge-Centric IoT—an architecture where data processing occurs closer to the source (e.g. sensors or edge devices) rather than relying solely on cloud computing. This enhances real-time decision-making, reduces dependence on internet connectivity and ensures faster MQA.

Smart sensors with inbuilt processing capabilities provide a streamlined solution, eliminating the need for additional hardware. Yet, their limited adoption (33%) indicates a need for further research and development to enhance their functionality and affordability. Meanwhile, cloud processing, though less frequently used, offers significant advantages for large-scale data integration, enabling centralised monitoring and advanced analytics. However, cloud bandwidth and latency issues persist, which need to be addressed for real-time analysis of MQA applications. The unique application of mobile-based processing demonstrates the potential for low-cost, user-friendly systems in small-scale dairy farms.

A hybrid strategy that integrates edge and cloud processing might provide optimal benefits, enabling immediate local real-time analysis for prompt decision-making while utilising cloud connectivity for enduring insights and scalability. Such systems would also facilitate the development of Federated Learning (FL) frameworks. FL is a decentralised ML approach that enables multiple nodes (edge processors) to collaboratively train a shared model without sharing raw data (Yurdem *et al*. 2024). In a hybrid edge-cloud set-up, FL allows local edge devices (e.g. on-site sensors or edge servers) to analyse milk quality in real time, detecting anomalies such as adulteration and spoilage. These local models periodically update a global model in the cloud, benefiting from aggregated insights without exposing sensitive data. This ensures scalability, robustness against data heterogeneity and compliance with data protection regulations, making milk quality monitoring more reliable and efficient.

Network layer

Several network layer technologies were utilised in the reviewed studies to enable data transmission within IoT systems. The distribution of communication protocols used in the studies is shown in Figure 2(c). Wi-Fi was featured in nine studies and was the most frequently used communication standard in the studies reviewed. The ESP8266 module served as a budget-friendly gateway linking hardware to cloud or onsite servers. Although the widespread availability and user-friendliness of Wi-Fi contribute to its popularity, its reliance on robust infrastructure may restrict its use on rural dairy farms. Ethernet was used in four studies, provided reliable, high-speed wired communication ideal for permanent setups. Global System for Mobile communication (GSM) was used in three studies. This facilitates SMS-based data transmission, offering farmers valuable insights such as fat content and milk volume on their deposited milk from the dairy processors. This technique is viable in regions that have inadequate Internet connectivity; however, it is becoming less relevant with the spread of mobile broadband. Bluetooth was used in two studies and ZigBee in one study, respectively, these methods support short-range communication, but their scalability is limited in large and broad IoT networks.

For data acquisition, serial communication protocols such as USB (utilised in 10 studies), I^2C and Serial Peripheral Interface (SPI) enabled efficient sensor-to-processor data transfer. RFID technology was used in three studies, improving traceability by recording milk quantity and farm details on tags, consistently using the EM18 module.

The reviewed studies indicated that traditional communication technologies such as Wi-Fi, GSM, USB, lack innovative solutions that are scalable, energy-efficient and suitable for a rural environment. There was a critical gap that included the absence of Low Range Wide Area Network (LoRaWAN) for long-range, low-power monitoring on remote dairy farms. Another gap was found in the underutilisation of next-generation internet (5G and 6G), for real-time spectral data transmission from sensors to processors. A combination of hybrid network strategies—LoRaWAN and 5G/6G—can provide long-distance transmission and ensure reliability. LoRaWAN can provide long-range low-power data communication for remote dairy farms, while cellular networks can be used where low-latency results are required, for example milk adulteration detection. If LoRaWAN connectivity drops due to interference or distance limitations, the system automatically switches to cellular, ensuring uninterrupted data flow.

Another research gap highlighted the lack of studies reporting on energy-efficient communication. Wi-Fi and GSM are power-intensive, often unsuitable for battery-operated sensors. The studies did not use energy harvesting from solar or RF, which could provide self-sustaining IoT nodes. This provides an opportunity for novel solutions such as using Wake-on-Radio (WOR) with technologies such as LoRaWAN, where sensors go into sleep mode until a gateway pings for data, extending the battery life of the sensors. Another low-powered communication that could be utilised is Bluetooth Low Energy (BLE), which could be useful for low-latency, short-distance communication. Future research must prioritise these advances to overcome infrastructure limitations, improve scalability and enable real-time monitoring of milk quality in diverse farming environments. These platforms provided remote access through web URLs and Android apps and were aimed primarily at consumers and dairy farmers. The remaining 21 studies employed custom Graphical User Interfaces (GUIs), which were designed to fit specific hardware and software setups. These GUIs provided both data visualisation capabilities and served as command hubs, enabling users to interact with the IoT system. However, most studies focussed on bespoke solutions, which come with significant drawbacks. When deployed across multiple locations or farms, these custom solutions often lack standardisation, making it difficult to scale effectively.

Service layer

Data processing for MQA using IoT technology showed variation across edge, cloud and hybrid edge-cloud methods. Of the 33 studies examined, 21 utilised edge processing on devices such as PCs, Raspberry Pi or an Arduino, indicating a preference for localised and immediate data processing, which is crucial in environments that demand rapid analysis with minimal delay. Despite its advantages, edge processing can be constrained by limited resources, affecting scalability and the ability to perform advanced data modelling. Cloud processing, though only observed in four studies, offered centralised storage and sophisticated data visualisation; however, its scalability heavily relies on consistent internet access, which poses challenges for remote dairy farms or areas with less developed infrastructure. The hybrid edge-cloud method, seen in eight studies, aimed to integrate the real-time processing strengths of edge computing with the enhanced data storage and visualisation capacities of the cloud. Numerous IoT commercial platforms shown in Figure 2(d), include ThingSpeak (four studies), BlynkIoT (two studies) and bespoke platforms (four studies), which are supported by graphical user interfaces, device handling and analytics. Conversely, platforms such as Amazon AWS and Google IoT were rarely employed, featuring in only one study each.

It is notable that 18 studies favoured bespoke solutions designed to fit their specific hardware and software requirements, but these often lacked common standards or commercial IoT platform support, resulting in limited interoperability and less accessible user interfaces. Furthermore, three studies used embedded systems with integrated sensors and displays for on-site monitoring, which provided instant outcomes but limited analytical capabilities.

Application layer

The application layer played a crucial role in facilitating data visualisation and user interaction in IoT-based systems for monitoring milk quality. Out of 12 studies, web-based dashboards and Android apps were utilised, leveraging SQL databases stored on servers or in the cloud for data visualisation and storage. The gap between web platforms and personalised GUIs highlights the contrasting goals in their system design. Web-based dashboards and apps provide enhanced flexibility and remote access but depend on internet connectivity, posing challenges in rural regions. Conversely, custom GUIs are more tailored to specific needs but might not offer the advanced features, user-friendliness and cross-platform compatibility found in commercial solutions.

A significant shortcoming identified in the reviewed studies is the insufficient use of advanced data analytics or visualisation tools, such as interactive charts, predictive models or AI-driven insights. For example, AI-powered anomaly detection could automatically indicate potential milk adulteration in real time, while predictive maintenance models could analyse sensor data to foresee equipment failures before they occur. These tools could substantially improve the functionality of the application layer, offering practical insights beyond basic visualisation and enabling proactive decision-making in the dairy supply chain. Future research should target the creation of hybrid solutions that mix the flexibility and accessibility of web-based platforms with the customisability of tailored GUIs, while ensuring they remain cost-effective, user-friendly and scalable.

AI used in MQA

Of the 43 studies analysed, 33 employed ML algorithms to address MQA challenges. This section outlines key considerations in the development of ML models across these studies, which represent the current state-of-the-art in MQA research. ML tasks were categorised into regression, classification and clustering. Specifically, 15 studies applied regression, 15 used classification, and one employed clustering. Two studies included both classification and regression tasks. The corresponding tasks and outcomes are summarised in Table 2, which supports the information presented in Assessment of technological integration: IoT and AI in milk quality analysis.

ML algorithms applied

The 33 reviewed papers report a total of 59 algorithm implementations across different categories. While some studies used multiple algorithms to identify the best fit for specific problems, Partial Least Squares Regression (PLSR) was the most widely applied algorithm overall, particularly for addressing issues related to milk composition analysis.

Figure 3 provides visual representations of the distribution of ML algorithms by category and their specific applications. Dimensionality reduction models and PLSR were dominant in milk composition analysis due to their ability to manage high-dimensional data effectively. Ensemble methods and regression-based models also featured prominently across studies for their robustness and adaptability to varying datasets. Artificial Neural Networks (ANNs) and other algorithms in the miscellaneous category were frequently utilised for detecting milk adulteration and microbial spoilage, as they could identify complex patterns in the data. Deep Learning (DL) models were primarily employed in multiclass classification tasks, such as identifying different types of adulteration. These were often integrated into IoT prototypes for proof-of-concept experiments, demonstrating their utility under real-world conditions.

The focus on dimensionality reduction methods such as principal component analysis (PCA) reflects their strength in handling large, multivariate datasets typical in milk quality studies. However, the choice of algorithm must align with the specific problem being addressed, as highlighted by the frequent use of ANN and DL models for adulteration detection and microbial spoilage identification. The success of these models under experimental conditions emphasises the importance of prototyping in IoT-enabled dairy systems. Without proper prototyping, there is a risk that models may not perform robustly in real-world conditions, leading to inaccurate predictions, unreliable performance or integration challenges with IoT hardware and sensor networks.

In order to maintain both accuracy and reliability, it is essential to update calibration models with live samples obtained from commercial sources (Diaz-Olivares *et al*. 2020). This step is crucial for adapting experimental findings to industrial-scale processes. Future research should focus on validating these algorithms under diverse environmental conditions and developing adaptive prototypes that can dynamically recalibrate based on real-time data from dairy farms. Establishing standardised evaluation protocols across studies will also facilitate comparisons and promote broader adoption of AI-based solutions in MQA applications.

Pre-processing techniques used

Various techniques were employed across studies as shown in Figure 4(b), with Principal Component Analysis (PCA) emerging as the most frequently used method. PCA proved particularly effective in milk composition analysis, where it handled large datasets by reducing dimensionality without significant loss of information. Savitzky–Golay (SG) smoothing, employed in six studies, was often paired with techniques such as standard normal variate or multivariate analysis such as Linear Discriminant Analysis (LDA) and multiple linear regression. This combination enhanced the interpretability of sensor signals, making them more suitable for AI/ML model training. For milk composition detection, Multiplicative Scatter Correction (MSC) presented mixed results.

While simplifying fat modelling, MSC negatively influenced protein content predictions due to its impact on slopes and offsets in multivariate analyses (Bogomolov *et al*. 2012). This limitation was especially pronounced when using regression techniques such as Partial Least Squares (PLS) regression, where MSC caused false predictions by amplifying the correlation between variance and analyte content (Bogomolov and Melenteva 2013).

Baseline correction techniques such as polynomial and linear baseline correction were primarily applied in studies involving analogue gas sensors, ensuring that raw sensor responses were appropriately normalised before being fed into AI/ML model for prediction.

The choice of pre-processing methods directly affects the performance of AI/ML models, with PCA and SG smoothing proving particularly versatile for handling large datasets and noisy sensor data. However, techniques such as MSC require careful consideration, as their impact on specific analytes (e.g. fat vs. protein) may vary. This highlights the importance of tailoring pre-processing methods to the unique characteristics of the dataset and application domain.

Future studies should prioritise systematic evaluations of pre-processing approaches, particularly in the context of multivariate regression models such as PLS. Furthermore, combining multiple pre-processing techniques could enhance prediction accuracy, provided their compatibility with the target dataset and downstream algorithms is validated. Standardising pre-processing pipelines across studies could also improve reproducibility and comparability of AI/ML applications in the dairy sector.

Model evaluation parameters

From the 33 studies incorporating AI/ML algorithms, 17 evaluation parameters were identified, as summarised in Figure 4(c). Many studies employed multiple metrics to assess model performance. Among these, Accuracy and Root Mean Squared Error (RMSE) emerged as the most commonly used, each appearing in 14 studies, followed by R^2 in 12 studies. Less frequently used metrics included Sensitivity,

Table 2 Summary of publications using AI algorithms in MQA.

Publication	Task	Model used	Result
Neto *et al.* (2019)	Detecting the presence of adulterants and predicting the type of adulterant used: Sucrose, starch, peroxide, bicarbonate, formaldehyde	Convolutional Neural Network (CNN), Random Forest (RF) and Gradient Boosting Machine (GBM) for comparison	CNN provided 98.76% classification accuracy for binary and 96.95% in the multi-class problem showing better performance than other tested methods
Amsaraj *et al.* (2021)	Classifying multiple adulterants in milk, simultaneously detecting adulterants—starch, urea, sucrose and quantifying the adulterants	Partial Least Squares—Discriminant Analysis (PLS-DA), ANN, Support Vector Machine (SVM)—(Support Vector regression (SVR) and Least Squares SVM (LS-SVM))	Classification accuracy of 100% was achieved by both SVM and PLS-DA models. For regression-based problems, SVM-based models performed better than other models since they could perform better with nonlinear data
Spieß *et al.* (2021)	To integrate untargeted spectra with clustering algorithms to identify atypical (deviating from normal milk fingerprint) milk characteristics and determine their underlying causes	GMM, K-means clustering	Could successfully provide meaningful categories of milk spectra that were atypical using clustering algorithms. However, more research is required to prove the hypothesis
Kumar *et al.* (2022)	Adulteration detection and milk grading using CNN	CNN	Was able to classify milk according to its quality and grade it accordingly
Bogomolov and Melenteva (2013)	The prediction of fats and protein content in milk based on light scattering principles for fats and proteins	Partial Least Squares Regression (PLSR)	Good prediction accuracy was achieved with $R^2 = 0.984$, Root Mean Squared Error (RMSE) = 0.07 and $R^2 = 0.965$ and RMSE = 0.04 respectively
Dixon *et al.* (2017)	Compression of MIR data and evaluation of nonlinearity between feature variables (MIR Spectral data) and milk quality parameters (Fats, Proteins and Lactose)	LS-SVM, Principal Component analysis (PCA)	Predictions made for proteins and fats showed nonlinear behaviours. To compress MIR data, nonlinear compression techniques are better than linear supervised compression techniques while using nonlinear predictive models for predictions due to their ability to map the nonlinear milk spectral data
Bogomolov *et al.* (2012)	Using PLSR for determining fat and protein content in milk based on the visible light scatter principle	PLSR	PLS Low prediction errors RMSE 0.05% and 0.03% were achieved for fats and proteins, respectively, using PLSR with VIS spectroscopy. Process in-line measurements can be conducted using this technique in dairy
Vimalajeewa *et al.* (2018)	Compression techniques are used to convert MIR milk quality data into compressed data domain	Wavelet transform, PCA, PLSR	These compression techniques could compress the data by 92.3% (PCA) and 91.1% (Wavelet transform) respectively, maintaining the same accuracy as pre-processed data. However, this technique can help improve computing capabilities when processing data on edge

(continued)

Table 2 (Continued).

Publication	Task	Model used	Result
Dina *et al*. (2019)	Regression problems targeting Fats, SNF, proteins and lactose were solved using Nonlinear regression models	SVR, Gaussian process Regression (GPR), PLSR, PCR	Nonlinear regression models such as GPR, and SVR are better for predicting milk-related data. SVR produced the best results for lactose and protein predictions, while GPR performed best for fat prediction in milk quality data
Lizhong *et al*. (2019)	Classification and evaluation of fermented milk based on aroma, taste and texture of milk	RF, Logistic Regression (LR) AdaBoosting	AdaBoosting model proposed by the authors in this work performed the best with a recognition accuracy of 96.8% as compared to RF and LR, which achieved 78.9% and 85.5%, respectively
Anand *et al*. (2019)	Development of Electronic Nose (E-Nose) for detection of adulterants—urea, formalin and hydrogen peroxide in milk using Linear Discriminant Analysis (LDA)	LDA	Classification accuracy of 90% was achieved
Natarajan and Vijayakumar (2021)	Development of Low-cost AI-based spectral sensor system to detect and classify five classes of adulterants in milk—sodium salicylate, Ammonium sulphate, hydrogen peroxide, dextrose and raw milk	LDA, SVM, Decision tree, Naive Bayes, Artificial Neural Network (ANN) and ANN with Genetic Algorithm	Accuracy of 92.7% was achieved with Neural network model which performed the best among all other models, a genetic algorithm was used as hyperparameter tuning of the neural network to improve the accuracy of the model to 100%
Moharkar and Patnaik (2019)	Detecting and quantifying the adulteration in milk IR laser-induced instrumentation	Multi-Layered Perceptron (MLP) neural network	Laser instrumentation proposed in the paper was able to classify the adulteration with 100% accuracy for urea, water and pure milk
Pablo *et al*. (2020)	Detection of antibiotics in milk using biosensor system with embedded machine learning algorithm for data processing	SVM.	The proposed solution outperformed state-of-the-art techniques for detecting antibiotics in milk in terms of speed and sensitivity.
Hosseini *et al*. (2021)	Detection of anionic surfactant in milk using NIR and ML-based classification	Interval partial least squares (iPLS), PLSR, PCA, Soft independent modelling of class analogies (SIMCA)	iPLS provided the best results compared to PLSR with SIMCA modelling. The models achieved an average sensitivity of 97.6%, specificity of 93% and, efficiency rate of 94.9% for adulterated samples while 100% classification accuracy for pure milk
Farah *et al*. (2021)	Adulteration detection of Bovine milk with Differential scanning calorimetry and machine learning techniques was proposed to detect whey, starch, urea and formaldehyde	RF, GBM and MLP	Good classification accuracy of 100% by GBM, MLP and 88.50% by RF was achieved, and the proposed solution of combining DSC with machine learning models can prove to be a promising tool for the dairy industry for quality monitoring and adulteration detection

(continued)

Table 2 (Continued).

Publication	Task	Model used	Result
Vasafi *et al.* (2018)	Detecting deviations in milk during milk processing procedures	PCA + regression, GPR and Autoencoder	PCA + Regression achieved an R^2 value of 0.7, while GPR achieved 0.97. However, GPR requires calibration with all possible changes before use. The autoencoder learns regular signal patterns during training and detects changes without calibration making it more reliable
Tohidi *et al.* (2018)	Detecting trace amounts of detergents in raw milk using artificial olfactory system	SVM, LDA, adaptive network-based fuzzy inference system (ANFIS)	nu-SVM provided with the best results achieving 90% accuracy for classification of different levels of adulterations in milk
Lal *et al.* (2022)	Detecting adulterants in milk by pH and electrical conductivity (EC) measurements using fuzzy analysis	Fuzzy Classification system	The pH and EC values of nonadulterated milk ranges from 6.45 to 6.67 and 4.56mS/cm to 5.26 ms/cm and pH increases with the addition of adulterants while the EC decreases
Vallejo *et al.* (2022)	Using Supervised learning with Wireless Power Transfer (WPT) technology to detect milk adulteration with different water concentrations	SVM, K Nearest Neighbour (KNN), Decision tree and LDA	Classification accuracy of 97.5% was achieved using supervised learning and WPT
Deshpande *et al.* (2021)	Classification and purity prediction of milk using machine learning	KNN	The solution successfully classified milk as either cow or buffalo with an accuracy of 92.45% and accurately estimated the percentage of water adulteration
Ady *et al.* (2018)	Using potentiometric, and gas sensor array with neural networks to detect microbial activity in milk	Back Propagation (BP) Neural Network	Identification rate of 83% was achieved with the mean square error of less than 0.0001
Patil *et al.* (2021)	Developing a low-cost system to determine data in milk using RF algorithm	RF	The study produced good reproducible and reliable results using the low-cost system proposed in the study
Fanglin *et al.* (2020)	Estimating the fats and proteins content in milk and classifying the milk based on the source of milk (dairy farm)	LR, SVM and RF are used for classification problem and Gradient Boosting Decision Tree (GBDT), Extreme gradient boosting (XGBT) and RF is used for estimating the content of fats and proteins in milk	SVM provided with the best results with accuracy of 95% for classification of milk source. At the same time, RF performed best for fats and protein estimations with $R^2 = 0.9399$ for fats and $R^2 = 0.9301$ for proteins
Diaz-Olivares *et al.* (2020)	Developing online on-farm milk composition analysis (fats, proteins and lactose) sensor system for real-time monitoring of milk	PLSR	RMSE values as small as 0.092% for fats and lactose and 0.110% for proteins, and R^2 value of 0.989, 0.894 and 0.689 was achieved for fats, proteins and lactose respectively for real-time predictions
Ma *et al.* (2018)	Using BP neural networks with fuzzy inference to monitor raw milk storage temperature	BP neural network	The relative error in predicting temperature was 4.88%

(continued)

Table 2 (Continued).

Publication	Task	Model used	Result
Muñiz *et al*. (2020)	Using handheld NIR sensor with the mobile application to predict fats, proteins, SNF and lactose concentrations in milk in real time	Neural network with adam algorithm	The study showed the application of milk quality testing in real-time on-site and new models could be added for different NIR sensors in the application and the mobile application could be deployed to other operating systems such as apples (iOs) providing a flexible and scalable solution
Yang *et al*. (2020a)	Developing on-site milk analyser for monitoring fats, proteins, lactose and total solids in real-time	PLSR	The study provided with RMSE values of 0.37%, 0.34%, 0.16% and 0.53% for fats, proteins lactose and total solids respectively and the correlation coefficient was 0.90, 0.57, 0.21 and 0.86 respectively for raw milk
Yang *et al*. (2020b)	Developing on-site portable detector to evaluate fat, proteins, lactose and total solids in raw and homogenised milk	PLSR	A compact detector with low measurement time was successfully designed which provided RMSE values of 0.14%, 0.14%, 0.08%, and 0.27% for raw milk and 0.10%, 0.12%, 0.08%, and 0.21% for homogenised milk for fats, proteins, lactose and total solids respectively
Prasanth *et al*. (2021)	Measuring intensity of specific wavelengths to determine and analyse fats and added water in milk	PLSR and PCA	Developing sensors to operate at specific wavelengths can help to eliminate complex algorithms and provide faster sensing applications
Muhammad *et al*. (2019)	Classification of protein content in milk using spectrophotometer sensor system	Kruskal–Wallis nonparametric test	Average error of 2.366% was achieved by the system, and the milk was classified into three protein content classes low, medium and high quality
Tao *et al*. (2022)	Infrared spectral sensor is used to analyse protein and fat content of milk using Gradient Boosting Regression Tree (GBRT)	GBRT	The study was able to achieve $R^2 = 0.996$ and $R^2 = 0.949$ for fats and proteins respectively and RMSE values of 0.085 and 0.058 respectively

F1-score, Mean Squared Error (MSE) and Mean Absolute Error (MAE), while specialised metrics such as Limit of Detection (LOD) and Recognition Capability appeared only once. The predominant use of accuracy and RMSE indicates a focus on general performance (i.e. robustness, interpretability and domain-specific knowledge) and error minimisation focussed specifically on reducing numerical discrepancies between predicted and actual values. Relying predominantly on these metrics can miss subtle elements of model assessment, such as sensitivity to false negatives, which is crucial for microbial quality indication, or the ability to interpret results in real-world contexts where adulterated milk is classified as pure milk. Additionally, the choice of metric often depended on the problem being addressed, suggesting variability in evaluation practices across studies.

While the identified evaluation metrics provide a broad spectrum for assessing model robustness, their application was inconsistent, with some studies neglecting to justify their choices. For example, metrics such as Area Under Curve (AUC) and specificity, critical for binary classification tasks, were underutilised despite their relevance in milk quality analysis. Similarly, advanced metrics addressing model reliability under varying conditions, such as robustness to noisy data, were rarely discussed.

Regarding validation methods, hold-out validation dominated, used in approximately 55% of the studies. Although

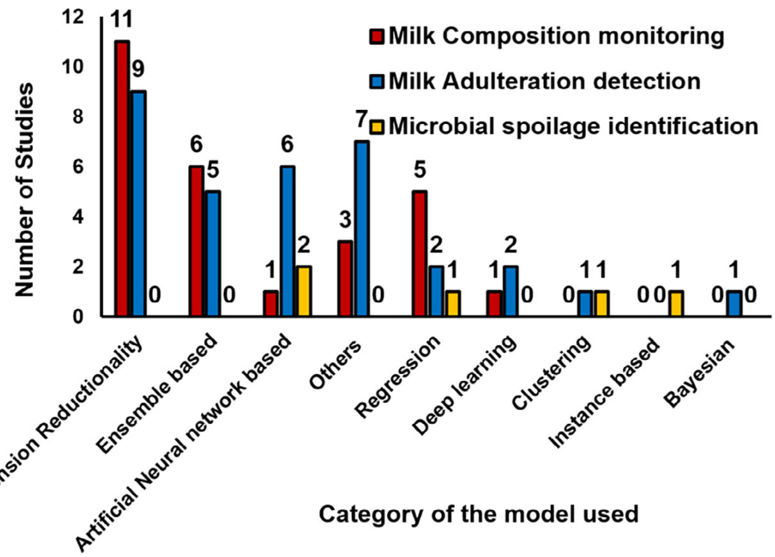


Figure 3 ML algorithms used per category.

straightforward, this approach may not adequately capture model generalisability, particularly with limited datasets. A more rigorous approach, such as k-fold cross-validation, was less commonly employed but offers a better assessment of model performance across diverse data subsets.

Future research should prioritise the systematic selection of evaluation metrics aligned with specific application needs. Additionally, adopting more robust validation methods and reporting standards such as clear metric justification, dataset transparency, and reproducibility protocols could enhance the comparability and reliability of results across studies, thereby advancing the field.

Recommended high-performance and widely adopted algorithms based on literature reviewed

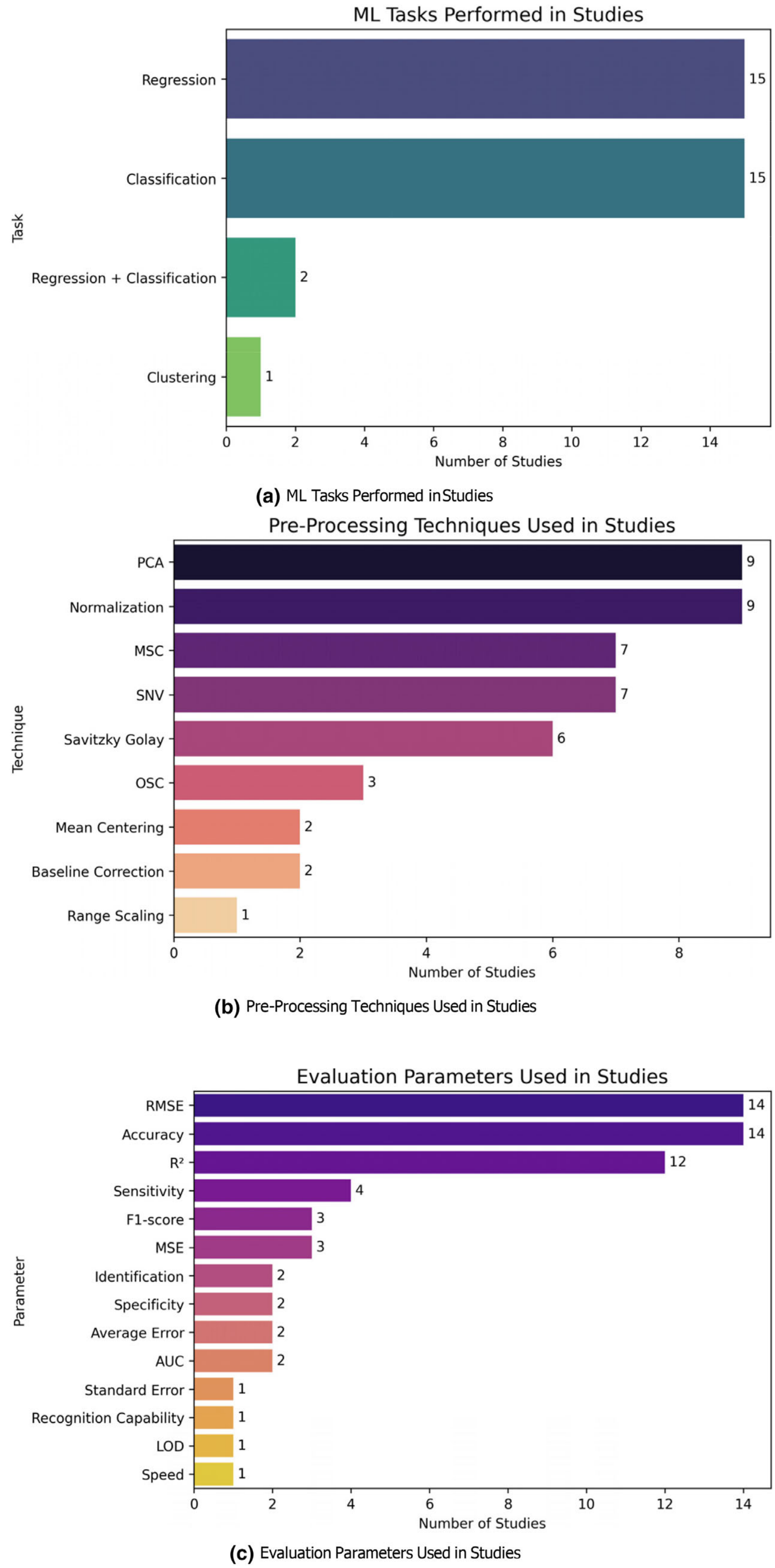
From the 33 studies analysed, dimensionality reduction algorithms emerged as the most applied techniques. However, determining the best-performing algorithms required focussing on 11 studies that compared multiple models. In these studies, the Support Vector Machine (SVM) was the best-performing algorithm in three cases (Tohidi *et al*. 2018; Fanglin *et al*. 2020; Amsaraj *et al*. 2021), demonstrating that it worked well with different spectral datasets. Similarly, Deep Learning (DL) (Vasafi *et al*. 2018; Neto *et al*. 2019) and Artificial Neural Networks (ANN) (Farah *et al*. 2021; Natarajan and Vijayakumar 2021) each performed the best twice, proving effective for complex, nonlinear data. Ensemble methods that combine multiple models to improve prediction accuracy by reducing bias and improving robustness, including AdaBoost were also highlighted twice for their strong predictive accuracy (Lizhong *et al*. 2019; Vallejo *et al*. 2022). While regression-based approaches were tested in four studies, only one study (Dina *et al*. 2019) found them superior, with nonlinear models such as Gaussian Process Regression (GPR) and Support Vector Regression (SVR) outperforming traditional regression techniques. This suggests a growing preference for algorithms capable of capturing nonlinear relationships present in milk spectral datasets.

The diversity of algorithm choices reflects the complexity of IoT data in dairy applications, where factors such as nonlinearity, high dimensionality, and data sparsity influence performance. The frequent selection of SVM highlights its robustness, but its computational complexity might limit scalability in real-time applications. DL and ANN, while powerful for large datasets, demand substantial computational resources, posing challenges for edge devices with limited capacity. Ensemble methods such as AdaBoost offer a balanced approach by enhancing prediction accuracy while maintaining efficiency, making them suitable for hybrid edge-cloud systems. Regression-based methods, though less favoured, could still prove valuable when paired with feature engineering and dimensionality reduction techniques.

Future research should explore the trade-offs between model accuracy and computational requirements, emphasising the scalability of high-performing algorithms in resource-constrained environments. Standardising evaluation criteria and conducting benchmark studies across diverse datasets would also improve the comparability of algorithmic performance.

Assessment of technological integration: IoT and AI in milk quality analysis

IoT significantly improves MQA by integrating a diverse range of sensors (spectral, gas, biosensors) into one system,



(a) ML Tasks Performed in Studies

(b) Pre-Processing Techniques Used in Studies

(c) Evaluation Parameters Used in Studies

Figure 4 Distribution of ML tasks, pre-processing techniques and evaluation parameters used in studies.

enabling continuous, real-time monitoring of milk quality parameters. IoT enables automation from data collection to processing and analysis of MQA applications, thereby reducing delays in detecting spoilage or adulteration of milk when compared to traditional lab-based methods. Edge computing combined with IoT provides low-latency analysis, while hybrid Edge-Cloud architectures can provide scalability, making it cost-effective for small-scale farms. Multi-sensor data fusion enhances accuracy by integrating various sensors onto one single system and also enables predictive insights into milk quality. IoT also ensures traceability using RFID and GSM, providing farmers with actionable alerts (e.g. enables farmers to monitor the protein and fat content in raw milk, helping them determine whether they are receiving a fair price for their product). However, challenges such as high-cost infrastructure, connectivity gaps in rural areas, lack of standardised protocols and lack of expertise limit their adoption.

AI has emerged as a powerful tool for analysing milk quality and detecting adulteration, leveraging spectroscopic techniques and advanced computational models. Various studies (Table 2) have demonstrated the effectiveness of ML in identifying milk adulterants, classifying milk species and assessing overall quality. The workflow for AI-Driven MQA is shown in Figure 5 based on the studies presented in this SLR.

AI enhances MQA by offering superior accuracy, efficiency and scalability compared to existing methods. Milk quality data, being high-dimensional spectral information, often includes both relevant and irrelevant features that can affect classical linear models negatively. Pre-processing is essential to remove noise but must be done carefully to retain crucial information. Linear models such as PCA and PLS struggle with nonlinear relationships in spectral data, while nonlinear ML methods such as SVM and ANN need dimensionality reduction and domain knowledge. DL offers a more effective alternative by automatically capturing complex patterns without extensive pre-processing or prior knowledge (Zhang *et al*. 2021).

Several research efforts have focussed on using spectral data, combined with ML algorithms. These approaches enable nondestructive and rapid detection of adulterants such as water, urea, formalin and synthetic milk. Supervised learning models, particularly SVM, RF and ANN, have shown high classification accuracy to distinguish pure milk from adulterated samples. The use of DL techniques, such as CNNs, has further enhanced predictive performance by extracting complex spectral features that traditional models might overlook.

AI models such as CNN, SVM and PLSR achieve over 95% accuracy in detecting adulterants such as starch, urea and hydrogen peroxide while also precisely predicting fat, protein and lactose content. Unlike manual testing, AI enables real-time, automated monitoring through portable sensors and edge computing, reducing reliance on costly lab measurements and minimising human error. Additionally, AI models can perform multi-task learning simultaneously, identifying adulteration, quantifying milk composition and classifying milk sources, making them more versatile than conventional approaches. Unsupervised learning methods, such as clustering and autoencoders, further improve detection by identifying unknown anomalies without predefined labels, ensuring adaptability to emerging threats.

A key trend observed in recent studies is the optimisation of ML models through dimensionality reduction and feature selection methods. PCA and Genetic Algorithms (GA) have been employed to refine spectral data inputs, improving computational efficiency while maintaining high detection accuracy. Additionally, hybrid models that integrate multiple classifiers or employ ensemble learning have demonstrated robustness in handling complex datasets with variations in milk composition.

ML-driven milk quality assessment has also been extended to species identification, ensuring the authenticity of dairy products. Techniques such as Partial Least Squares Discriminant Analysis (PLS-DA) and k-Nearest Neighbours (k-NN) have been used to differentiate milk from various animal sources, thereby addressing issues related to mislabelling, authenticity and transparency.

Despite the significant advancements, challenges remain in developing generalised models applicable across diverse spectral milk samples. Factors such as regional variations in milk composition, differences in adulterant concentrations and the need for real-time processing necessitate further research into model adaptability and transfer learning techniques. AI could also play a vital role for AI-optimised networking to balance latency, power consumption and bandwidth. For instance, ML models could predict the best communication protocol to be used for data transmission (i.e. 5G for spectral images of milk, LoRaWAN for temperature) based on the available resources. Future work may focus on federated learning approaches to enhance model generalisation while maintaining data privacy for MQA applications. Overall, ML-based methods have revolutionised milk quality analysis by providing accurate, rapid and scalable solutions for detecting adulteration and ensuring authenticity. Continued advancements in spectroscopic sensing technologies and computational techniques will further enhance the reliability and applicability of these technologies in the dairy industry.

The integration of IoT and AI within the dairy industry facilitates rapid, cost-efficient and highly accurate quality control by enabling real-time monitoring and analysis of critical parameters. This advanced approach supports data-driven decision-making across the supply chain, ensuring optimal product quality, traceability and compliance with industry standards.

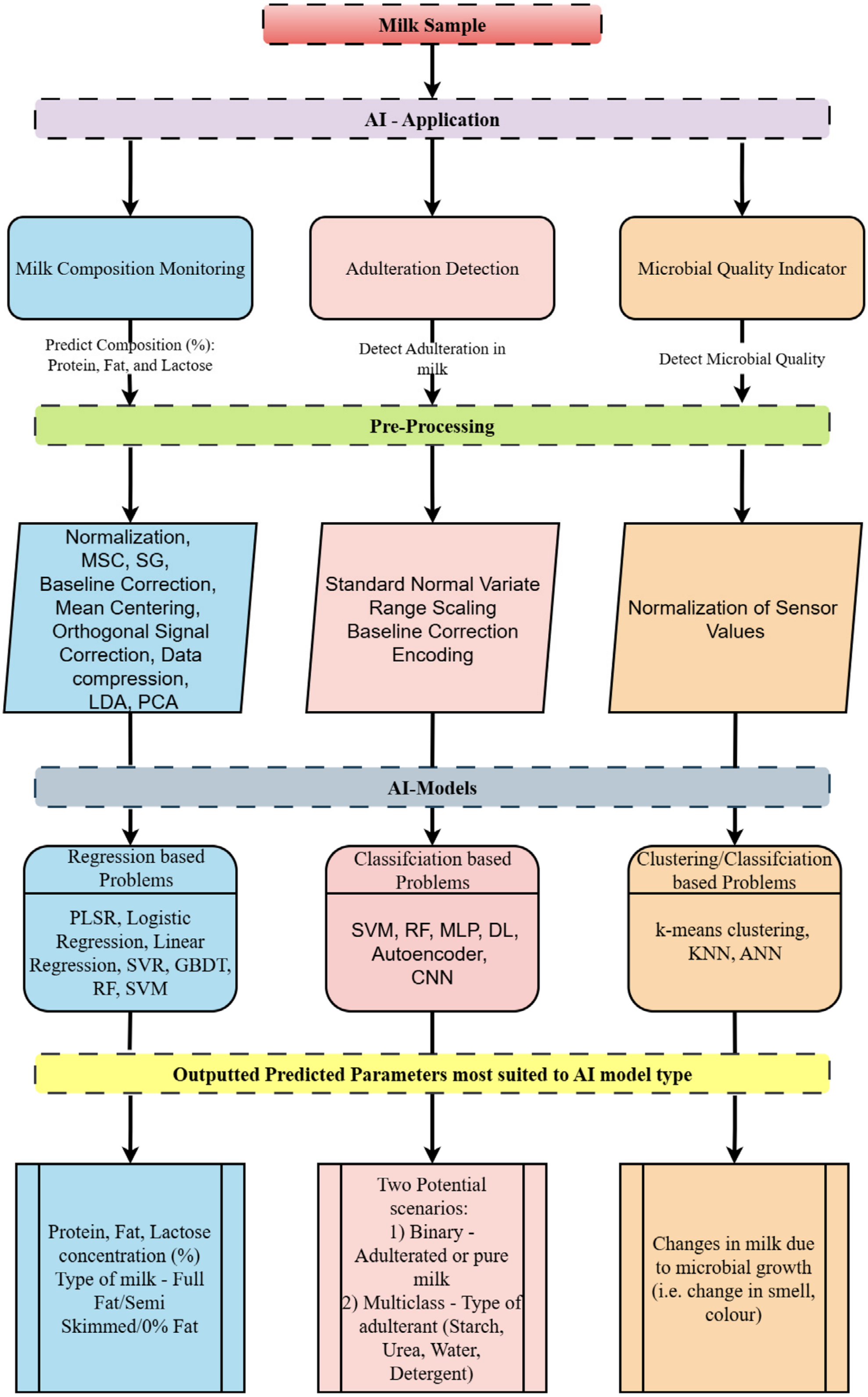


Figure 5 Artificial intelligence (AI)-driven milk quality analysis and monitoring workflow – colour of workflow indicates AI pathway for milk composition (blue), adulteration detection (red), microbial quality indicators (orange).

CONCLUSION

Milk Quality Analysis was the primary focus of this study, whereby a systematic literature review was generated from 43 studies published over the past decade. The three main subcategories covered were milk composition monitoring, milk adulteration detection and microbial quality indicators in milk using IoT and AI. The objective was to present the current state-of-the-art in IoT architecture implementation and AI application in dairy production (MQA).

Our findings indicate a growing demand for real-time measurements in the dairy industry, driven by the increasing implementation of IoT and digitisation. Spectroscopy emerged as a predominant solution (62%) for providing predictions for compositional constituents and detecting adulteration. Bacterial spoilage identification in milk was addressed using gas sensors instead of spectroscopy. In terms of processing, 64% of the studies utilised edge processing, while 36% incorporated cloud computing, with only 12% processing data over the cloud. Developing hybrid edge-cloud platforms could prove beneficial for the dairy industry. However, challenges such as high computational requirements for raw spectral data transmission and the limitations of Wi-Fi technology need to be addressed. Exploring wireless technologies such as BLE and ZigBee are examples that could enhance data communication.

Data pre-processing often involved PCA for spectral data, contributing to improved outputs for AI models. Dimensionality reduction algorithms were prevalent in more than half of the studies employing ML algorithms, with Ensemble and Other category algorithms shown to perform the best. SVM and RF algorithms were notable choices for robust predictions in MQA. The DL approach demonstrated effectiveness in classification-related problems, particularly in milk adulteration detection, requiring further exploration. Combining multiple ML algorithms can enhance efficiency and robustness in milk quality predictions. Addressing challenges related to data size, hardware errors and insufficient training data is crucial to prevent inaccurate predictions. Providing open access to milk composition and quality datasets for researchers can help address overfitting by enabling diverse model training and facilitating the establishment of robust benchmarks.

The integration of advanced sensors, optimisation of hybrid edge-cloud platforms, and the adoption of 5G technology will improve real-time data collection and processing. Advancements in AI models, with a focus on DL, will further enhance decision-making processes. DL models can automatically learn patterns and features from large milk spectral data, improving predictive accuracy.

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AUTHOR CONTRIBUTIONS

Rahul Mhapsekar: Conceptualization; formal analysis; investigation; methodology; writing – original draft. **Deirdre Kilbane:** Supervision; writing – review and editing. **Steven Davy:** Supervision; writing – review and editing. **Lizy Abraham:** Supervision; writing – review and editing. **Mark Fenelon:** Conceptualization; writing – review and editing. **Norah O'Shea:** Conceptualization; supervision; writing – review and editing.

CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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None reported.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during this study.

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